

Exercise 1a Eigenstates of spin-1/2 in magnetic field.

$\mathcal{H} = -\gamma \vec{B} \cdot \vec{S}$. Orient axes so that $\vec{B} = B\hat{z}$. Then, $\mathcal{H} = -\gamma B S_z \implies$ stationary states are eigenstates of S_z , i.e $|\uparrow\rangle, |\downarrow\rangle$ with energies $\mathcal{H}|\uparrow\rangle = -\frac{\gamma\hbar B}{2}|\uparrow\rangle$ and $\mathcal{H}|\downarrow\rangle = \frac{\gamma\hbar B}{2}|\downarrow\rangle$.

Exercise 1c Spin 1/2 system in magnetic field $\vec{B} = (B_x, 0, B_z)$.

Expressing $\mathcal{H}$ in matrix form w.r.t S_z basis gives

$$\mathcal{H} = -\frac{\gamma\hbar}{2} \left(B_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + B_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) = -\frac{\gamma\hbar}{2} \begin{pmatrix} B_z & B_x \\ B_x & -B_z \end{pmatrix}$$

Need to solve IVP

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = \mathcal{H}|\psi(t)\rangle, \quad |\psi(0)\rangle = |\uparrow\rangle$$

Express $|\psi(t)\rangle = \begin{pmatrix} a(t) \\ b(t) \end{pmatrix}$ so that we have the system of ODEs

$$\begin{aligned} \dot{a} &= \hat{\gamma}(B_z a + B_x b), & a(0) &= 1 \\ \dot{b} &= \hat{\gamma}(B_x a - B_z b), & b(0) &= 0 \end{aligned}$$

where $\hat{\gamma} = i\gamma/2$. Differentiating again yields

$$\begin{aligned} \ddot{a} &= \hat{\gamma} B_z \dot{a} + \hat{\gamma} B_x \dot{b} \\ &= \hat{\gamma} B_z (\hat{\gamma} B_z a + \hat{\gamma} B_x b) + \hat{\gamma} B_x (\hat{\gamma} B_x a - \hat{\gamma} B_z b) \\ &= \hat{\gamma}^2 |B|^2 a \\ &= -\frac{\gamma^2}{4} |B|^2 a \end{aligned}$$

where $|B|^2 = B_x^2 + B_z^2$. Similarly, $\ddot{b} = -\frac{\gamma^2}{4} |B|^2 b$. These are SHM equations with solutions,

$$\begin{aligned} a(t) &= C \cos\left(\frac{\gamma|B|t}{2}\right) + D \sin\left(\frac{\gamma|B|t}{2}\right) \\ b(t) &= E \cos\left(\frac{\gamma|B|t}{2}\right) + F \sin\left(\frac{\gamma|B|t}{2}\right) \end{aligned}$$

Plugging in I.Cs gives

$$\begin{aligned} a(0) &= C = 1 \\ b(0) &= E = 0 \\ \dot{a}(0) &= \hat{\gamma} B_z = D \frac{\gamma|B|}{2} \implies D = \frac{2\hat{\gamma} B_z}{\gamma|B|} = i \frac{B_z}{|B|} \\ \dot{b}(0) &= \hat{\gamma} B_x = F \frac{\gamma|B|}{2} \implies F = \frac{2\hat{\gamma} B_x}{\gamma|B|} = i \frac{B_x}{|B|} \end{aligned}$$

So, the solution is

$$|\psi(t)\rangle = \begin{pmatrix} \cos\left(\frac{\gamma|B|t}{2}\right) + \frac{iB_z}{|B|} \sin\left(\frac{\gamma|B|t}{2}\right) \\ \frac{iB_x}{|B|} \sin\left(\frac{\gamma|B|t}{2}\right) \end{pmatrix}$$

Then, the probability of measuring spin up at time $t = \tau$ is

$$\begin{aligned} |a(\tau)|^2 &= \left| \cos\left(\frac{\gamma|B|\tau}{2}\right) + \frac{iB_z}{|B|} \sin\left(\frac{\gamma|B|\tau}{2}\right) \right|^2 \\ &= \cos^2\left(\frac{\gamma|B|\tau}{2}\right) + \frac{B_z^2}{|B|^2} \sin^2\left(\frac{\gamma|B|\tau}{2}\right) \end{aligned}$$

Note $B_x \rightarrow 0 \implies |B|^2 \rightarrow B_z^2 \implies |a(\tau)|^2 \rightarrow 1$.

Exercise 2a Write down the (time-dependent) Hamiltonian for the spin-1/2 system

$$\mathcal{H} = -\gamma \mathbf{S} \cdot \mathbf{B} = -\gamma[S_z B_0 + S_x B_1 \cos \nu t]$$

Exercise 2b Derive ODE for system expressing $|\chi\rangle = a(t)|0\rangle + b(t)|1\rangle$ where $S_z|0\rangle = \hbar/2|0\rangle$ and $S_z|1\rangle = -\hbar/2|1\rangle$.

Have

$$\begin{aligned} \mathcal{H}|\chi\rangle &= -\frac{\gamma\hbar}{2}(a(t)B_0|0\rangle - b(t)B_0|1\rangle + a(t)B_1 \cos(\nu t)|1\rangle + b(t)B_1 \cos(\nu t)|0\rangle) \\ &= -\frac{\gamma\hbar}{2}((a(t)B_0 + b(t)B_1 \cos(\nu t))|0\rangle + (a(t)B_1 \cos(\nu t) - b(t)B_0)|1\rangle) \end{aligned}$$

Schrödinger gives $\mathcal{H}|\chi\rangle = i\hbar \frac{d|\chi\rangle}{dt} = \dot{a}(t)|0\rangle + \dot{b}(t)|1\rangle$. Setting equal to above and using orthonormality of basis, can decompose into

$$\dot{a}(t) = i\frac{\omega}{2}a(t) + i\Omega \cos(\nu t)b(t) \quad (1)$$

$$\dot{b}(t) = -i\frac{\omega}{2}b(t) + i\Omega \cos(\nu t)a(t) \quad (2)$$

with

$$\omega = \gamma B_0$$

$$\Omega = \frac{1}{2}\gamma B_1$$

Exercise 2c Derive using rotating frame $a(t) = \alpha(t)e^{i\nu t/2}, b(t) = \beta(t)e^{-i\nu t/2}$ and Rotating Wave Approximation ($\nu \gg \omega, \Omega$).

$$\dot{\alpha}(t) = i\frac{\delta}{2}\alpha(t) + i\frac{\Omega}{2}\beta(t) \quad (3)$$

$$\dot{\beta}(t) = -i\frac{\delta}{2}\beta(t) + i\frac{\Omega}{2}\alpha(t) \quad (4)$$

Exercise 2d Solve system with IC $|\chi(0)\rangle = |0\rangle$

Equivalent to 1C, get

$$\begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{1}{2}\Omega_R t\right) + i\frac{\delta}{\Omega_R} \sin\left(\frac{1}{2}\Omega_R t\right) \\ i\frac{\Omega}{\Omega_R} \sin\left(\frac{1}{2}\Omega_R t\right) \end{pmatrix}$$

Exercise 2e Solve system at resonance $\delta = 0$ with $\Omega(t)$ time-dependent.

$$\begin{aligned}\dot{\alpha}(t) &= \frac{i\Omega(t)}{2}\beta(t) \\ \dot{\beta}(t) &= \frac{i\Omega(t)}{2}\alpha(t)\end{aligned}$$

Set $f = \alpha - \beta$ and $g = \alpha + \beta$. Then, we can express the system as

$$\begin{aligned}\dot{f} &= -\frac{i\Omega(t)}{2}f \\ \dot{g} &= \frac{i\Omega(t)}{2}g\end{aligned}$$

Using $A(t) = \int^t \Omega(\tau) d\tau$, separating variables and solving gives

$$f(t) = Ce^{-iA(t)/2}, \quad g(t) = De^{iA(t)/2}$$

I.Cs give $f(0) = \alpha(0) - \beta(0) = 1$ and $g(0) = \alpha(0) + \beta(0) = 1$. So, $C = D = 1$. Returning to α, β , we have

$$\begin{aligned}\alpha(t) &= \frac{1}{2}(f(t) + g(t)) = \frac{1}{2}(e^{-iA(t)/2} + e^{iA(t)/2}) = \cos\left(\frac{A(t)}{2}\right) \\ \beta(t) &= \frac{1}{2}(g(t) - f(t)) = \frac{1}{2}(e^{iA(t)/2} - e^{-iA(t)/2}) = i \sin\left(\frac{A(t)}{2}\right)\end{aligned}$$

Exercise 2g Write Hamiltonian for RWA ODE system

$$i\hbar \frac{d}{dt} \begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} \delta & \Omega \\ \Omega & -\delta \end{pmatrix} \begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix}$$

Exercise 2h Eigenstates of $\hat{\mathbf{H}}_{RWA}$? Eigenstates when $\Omega \gg \delta$ and $\delta \gg \Omega$? Plot eigenenergies for fixed Ω . Energy splitting at $\delta = 0$?

Eigenstates found identically to question 1. Get

$$|E_{\pm}\rangle = \begin{pmatrix} \delta \pm \sqrt{\delta^2 + \Omega^2} \\ \Omega \end{pmatrix}, \quad E_{\pm} = \pm \frac{1}{2} \sqrt{\delta^2 + \Omega^2}$$

For $\delta \gg \Omega$, hamiltonian looks diagonal so expect basis states to be eigenstates. For $\Omega \gg \delta$, hamiltonian looks like σ_x so expect symmetric and antisymmetric combinations of basis states to be eigenstates.

Splitting at resonance is given by $E_+(\delta = 0) - E_-(\delta = 0) = \sqrt{\Omega^2} - (-\sqrt{\Omega^2}) = \Omega$

Exercise 2i How does this entire analysis change if $\vec{B}_1 \parallel \hat{y}$? Determine $\mathbf{H}_{RWA}$ for $\vec{B}_{AC} = B_1(\cos \theta, \sin \theta, 0) \cos \nu t$. You should find it behaves exactly the same as if $\vec{B}_{AC} = B_1(1, 0, 0) \cos(\hat{t} - \theta)$!

Proof. Recalling $S_y|0\rangle = i\frac{\hbar}{2}|1\rangle$ and $S_y|1\rangle = -i\frac{\hbar}{2}|0\rangle$, we see that

$$\begin{aligned}\mathcal{H}|\chi\rangle &= -\frac{\gamma\hbar}{2}(a(t)B_0|0\rangle - b(t)B_0|1\rangle + a(t)B_1\cos(\nu t)|1\rangle + b(t)B_1\cos(\nu t)|0\rangle) \\ &= -\frac{\gamma\hbar}{2}((a(t)B_0 + b(t)B_1\cos(\nu t))|0\rangle + (a(t)B_1\cos(\nu t) - b(t)B_0)|1\rangle)\end{aligned}$$

Setting up as before, we find

$$\begin{aligned}\dot{a}(t) &= \frac{i\omega}{2}a(t) + i\Omega\cos(\nu t)e^{-i\theta}b(t) \\ \dot{b}(t) &= -\frac{i\omega}{2}b(t) + i\Omega\cos(\nu t)e^{-i\theta}a(t)\end{aligned}$$

We can proceed as before in the rotating frame using $\alpha(t) = a(t)e^{i\nu t/2}$ and $\beta(t) = b(t)e^{-i\nu t/2}$ and get the equations

$$\begin{aligned}\dot{\alpha}(t) &= \frac{i\delta}{2}\alpha(t) + i\frac{\Omega}{2}(1 + e^{-2i\nu t})e^{-i\theta}\beta(t) \\ \dot{\beta}(t) &= -\frac{i\delta}{2}\beta(t) + i\frac{\Omega}{2}(1 + e^{2i\nu t})e^{i\theta}\alpha(t)\end{aligned}$$

Applying the rotating wave approximation makes the $e^{\pm 2i\nu t} \rightarrow 0$, so our Hamiltonian looks like

$$\mathbf{H}_{RWA} = -\frac{\hbar}{2} \begin{pmatrix} \delta & \Omega e^{-i\theta} \\ \Omega e^{i\theta} & -\delta \end{pmatrix}$$

If instead we used $B_{AC} = B_1(1, 0, 0) \cos(\nu t - \theta)$, we would arrive at

$$\begin{aligned}\dot{a}(t) &= i\frac{\omega}{2}a(t) + i\Omega\cos(\nu t - \theta)b(t) \\ \dot{b}(t) &= -i\frac{\omega}{2}b(t) + i\Omega\cos(\nu t - \theta)a(t)\end{aligned}$$

When we move to the rotating frame in the first equation,

$$\cos(\nu t - \theta) = \frac{1}{2}(e^{-i\theta}e^{i\nu t} + e^{i\theta}e^{-i\nu t}) \rightarrow \frac{1}{2}(e^{-i\theta}e^{i\nu t} + e^{i\theta}e^{-i\nu t})e^{-i\nu t} = \frac{1}{2}(e^{-i\theta} + e^{i\theta}e^{-2i\nu t})$$

The RWA removes the latter term so that the same phase factor $e^{-i\theta}$ emerges. Similarly, we get the factor $e^{i\theta}$ in the bottom equation. $\square$

Exercise 2j What happens if $\mathbf{B}_1 \parallel \hat{z}$

Then our hamiltonian just looks like $\mathcal{H} = -\gamma S_z(B_0 + B_1 \cos \nu t)$ and we get zeeman splitting modulated by $B_1 \cos \nu t$.

$$E_{\pm} = \mp \gamma \frac{\hbar}{2} (B_0 + B_1 \cos \nu t)$$

Exercise 3a**Exercise 3b**

(i)

$$\hat{U}_{\pi_x/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$

We see that $\hat{U}_{\pi/2}$ takes $|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |0_y\rangle$ and $|1\rangle \rightarrow \frac{1}{\sqrt{2}}(i|0\rangle + |1\rangle) = |1_y\rangle$. It rotates the Bloch vector by $\pi/2$ counter clockwise around the x -axis.

(ii)

$$\hat{U}_{\pi_x} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

Rotates around x axis by π , takes $|0\rangle \leftrightarrow |1\rangle$

Exercise 3c

(i)

$$\hat{U}_{\pi_y/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Takes $|0\rangle \rightarrow |1_x\rangle$ and $|1\rangle \rightarrow |0_x\rangle$. Rotates by $\pi/2$ ccw around y axis

(ii)

$$\hat{U}_{\pi_y} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

π turn around y axis. Flips $|0\rangle \leftrightarrow |1\rangle$

Exercise 3d

Then, Note

$$\hat{U}(\Omega = 0, \delta, t) = \begin{pmatrix} e^{i\delta t/2} & 0 \\ 0 & e^{-i\delta t/2} \end{pmatrix}$$

$$\begin{aligned} \hat{U}_{\text{Ramsey}} &= \hat{U}_{\pi_x/2} \hat{U}(0, 0, t) \hat{U}_{\pi_x/2} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} e^{i\delta t/2} & 0 \\ 0 & e^{-i\delta t/2} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} e^{i\delta t/2} & i e^{i\delta t/2} \\ i e^{-i\delta t/2} & e^{-i\delta t/2} \end{pmatrix} \\ &= \begin{pmatrix} i \sin(\delta t/2) & i \cos(\delta t/2) \\ i \cos(\delta t/2) & -i \sin(\delta t/2) \end{pmatrix} \end{aligned}$$

If state starts in $|0\rangle$, then after Ramsey Sequence $|0\rangle \rightarrow i \sin(\delta t/2)|0\rangle + i \cos(\delta t/2)|1\rangle$ so there is probability $\cos^2(\delta t/2)$ that it is in $|1\rangle$.

If we apply a $\pi_y/2$ resonance pulse last instead, we get

$$\begin{aligned}\hat{U} &= \hat{U}_{\pi_y/2} \hat{U}(0, 0, t) \hat{U}_{\pi_x/2} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{i\delta t/2} & 0 \\ 0 & e^{-i\delta t/2} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{i\delta t/2} & ie^{i\delta t/2} \\ ie^{-i\delta t/2} & e^{-i\delta t/2} \end{pmatrix}\end{aligned}$$

The relevant matrix element for the transition amplitude $|0\rangle \rightarrow |1\rangle$ is the lower left, which is given by

$$\frac{1}{2}(-e^{i\delta t/2} + ie^{-i\delta t/2})$$

The probability is then

$$\begin{aligned}P &= \left| \frac{1}{2}(-\cos(\delta t/2) - i \sin(\delta t/2) + i \cos(\delta t/2) + \sin(\delta t/2)) \right|^2 \\ &= \frac{1}{4}((\sin \delta t/2 - \cos(\delta t/2))^2 + (\sin \delta t/2 - \cos \delta t/2)^2) \\ &= \frac{1}{2}(1 - 2 \sin(\delta t/2) \cos(\delta t/2)) \\ &= \frac{1}{2}(1 - \sin(\delta t))\end{aligned}$$